

# MRISB 4-STEP LESSON DESIGN | UbD × ETEC

**OUTCOME → PROOF → PREPARE → RESPOND → REFLECT** Plan the learning first; let the checklist follow. Click a field to type • click a box to tick it.

| Teacher                                                                                                                                                                                                             | Subject                           | Grade / Section                                                                                                                                                                                                                                                               | Date / Period                     | Unit / Topic                              |
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| Mr. Ashraf Elbialy                                                                                                                                                                                                  | Computer Studies (DigiChamps L10) | Grade 10 - Classes 10/1 to 10/6                                                                                                                                                                                                                                               | Wk 13 (15-19 Nov 2026) - Period 1 | Unit 4 - 4.2 Core Logic of the Game (1/2) |
| <b>Pacing (2-1-1-2):</b> <input checked="" type="checkbox"/> On plan <input type="checkbox"/> Ahead <input type="checkbox"/> Behind<br><b>Weeks ahead / behind:</b> 0 <b>Week / lesson:</b> Week 13 / Lesson 1 of 2 |                                   | <b>Starting point (2-2-1-3):</b> What did my last check (quiz, exit ticket, homework) show — and what am I changing today because of it?<br>Countdowns working for ~80%; ms conversion now secure. Today: start the game, make worms pop out at random times in random holes. |                                   |                                           |

## 1 OUTCOMES — What must students learn?

**Desired learning outcome(s)** — trace to the curriculum / standard and the table of specifications (ToS)

Students build core game logic: start the game, make worms pop out, set random times (Math.random range formula) and choose random holes. Sprint 4.2 obj. 1-4; CSTA 3A-AP-16; Activity 1.

**Success criteria** — what will successful learning look / sound like?

I start the game on a button click; I generate a random time between min and max; I choose a random hole; worms pop out repeatedly.

**Key knowledge / skill / understanding:**

Math.random() gives 0 to <1; min + rand\*(max-min); Math.floor for whole numbers; array of holes.

**Likely misconception / access or challenge need:**

Math.random gives whole numbers; same worm can't appear twice in the same hole.

## 2 PROOF — What will students do to show they have learned it? Three activities, each with a DIFFERENT strategy, resource and evidence tool. Then tick the Coverage Check on page 2.

| ACTIVITY 1 student task + evidence of learning                                                                                                                                                                                                                                                                                    | ACTIVITY 2 student task + evidence of learning                                                                                                                                                                                                                                                                                                | ACTIVITY 3 student task + evidence of learning                                                                                                                                                                                                                                                                                    |
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| <b>Strategy:</b> Problem solving <input checked="" type="checkbox"/> <b>Time:</b> 8 min                                                                                                                                                                                                                                           | <b>Strategy:</b> Modelling / worked exam <input checked="" type="checkbox"/> <b>Time:</b> 22 min                                                                                                                                                                                                                                              | <b>Strategy:</b> Debate <input checked="" type="checkbox"/> <b>Time:</b> 7 min                                                                                                                                                                                                                                                    |
| <b>What students will do:</b><br>Activity 1: calculate a random time between 2000 and 3000 ms for rand = 0.83 by hand, then verify in Chrome's Console.                                                                                                                                                                           | <b>What students will do:</b><br>Code along: startGame(), randomTime(), randomHole() and popOut() using setTimeout; worms appear in random holes until the game ends.                                                                                                                                                                         | <b>What students will do:</b><br>Is a game fair if it is random? Teams argue for/against using real examples (dice, sports draws) and suggest how to make randomness feel fair.                                                                                                                                                   |
| <b>Evidence tool:</b> Mini-whiteboards <input checked="" type="checkbox"/><br>Assessment: <input checked="" type="checkbox"/> Diagnostic <input type="checkbox"/> Formative <input type="checkbox"/> End<br>Grouping: <input checked="" type="checkbox"/> Individual <input type="checkbox"/> Pair <input type="checkbox"/> Group | <b>Evidence tool:</b> Practical / performance task <input checked="" type="checkbox"/><br>Assessment: <input type="checkbox"/> Diagnostic <input checked="" type="checkbox"/> Formative <input type="checkbox"/> End<br>Grouping: <input checked="" type="checkbox"/> Individual <input type="checkbox"/> Pair <input type="checkbox"/> Group | <b>Evidence tool:</b> Oral questioning <input checked="" type="checkbox"/><br>Assessment: <input type="checkbox"/> Diagnostic <input checked="" type="checkbox"/> Formative <input type="checkbox"/> End<br>Grouping: <input type="checkbox"/> Individual <input type="checkbox"/> Pair <input checked="" type="checkbox"/> Group |
| <b>Resources:</b><br>Chrome Console; formula slide.                                                                                                                                                                                                                                                                               | <b>Resources:</b><br>VS Code; project source 'Core logic'; student book 4.2.                                                                                                                                                                                                                                                                  | <b>Resources:</b><br>Debate prompt; examples.                                                                                                                                                                                                                                                                                     |
| <b>Support (low performers / SEN):</b><br>Calculator; worked example with 0.5.                                                                                                                                                                                                                                                    | <b>Support (low performers / SEN):</b><br>Code skeleton with functions named; pair programming.                                                                                                                                                                                                                                               | <b>Support (low performers / SEN):</b><br>Sentence starters.                                                                                                                                                                                                                                                                      |
| <b>Stretch (fast learners / gifted):</b><br>Write a function randomBetween(min,max) and test 5 values.                                                                                                                                                                                                                            | <b>Stretch (fast learners / gifted):</b><br>Prevent the same hole twice in a row (compare with lastHole).                                                                                                                                                                                                                                     | <b>Stretch (fast learners / gifted):</b><br>Link to probability: chance of the same hole twice with 9 holes.                                                                                                                                                                                                                      |

| 3 PREPARE — What do I need to do so students can succeed?                                                                                                                                                                                                                                                                                                                                                                                                                                          | 4 RESPOND — What will I do with what I see and hear?                                                                                                                                                                                                                                                                   |
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| <b>Set students up to succeed:</b><br><input checked="" type="checkbox"/> Hook / prior learning <input checked="" type="checkbox"/> Explain or model <input checked="" type="checkbox"/> Success criteria shared<br><input checked="" type="checkbox"/> Key vocabulary / examples <input checked="" type="checkbox"/> Real-life connection <input checked="" type="checkbox"/> Clear instructions / resources<br><input checked="" type="checkbox"/> Praise, recognition and reinforcement planned | <b>If students struggle or finish early I will:</b><br><input checked="" type="checkbox"/> Re-teach <input checked="" type="checkbox"/> Regroup <input checked="" type="checkbox"/> Give extra support<br><input checked="" type="checkbox"/> Stretch <input checked="" type="checkbox"/> Have students redo / improve |
| <b>My setup / modelling:</b> Time: 6   min<br>Hook: roll a dice - how can a computer 'roll'? Model Math.random, the range formula, Math.floor with an array of holes. Show the flow: start -> random hole -> pop out -> hide -> repeat. Share success criteria.                                                                                                                                                                                                                                    | <b>What I will watch for / likely responses:</b><br>Using Math.round giving index 9 (out of range); setTimeout not called again; CSS class for 'up' worm missing. Likely: 'worms appear all at once'.                                                                                                                  |
| <b>What is new or innovative in this lesson?</b> (ETEC looks for this)<br>Numeracy and probability embedded in coding; students debate fairness of randomness, linking maths to game design ethics.                                                                                                                                                                                                                                                                                                | <b>Close / exit check</b> (summative + time for learners' questions)   Time: 7   min<br>Exit ticket: compute min + 0.25*(max-min) for 1000-4000; why Math.floor not Math.round? Learners' questions.                                                                                                                   |
| <b>Progress recorded / shared:</b><br><input checked="" type="checkbox"/> Gradebook <input type="checkbox"/> Parents <input type="checkbox"/> Written feedback in notebook / portfolio                                                                                                                                                                                                                                                                                                             | <b>Homework / platform</b> (e.g. Madrasati)<br>Madrasati: finish pop-out logic; upload a 20-second screen recording or screenshot. Optional: Challenge 3 Start/Stop random colour.                                                                                                                                     |

| 5 REFLECT — After the lesson <i>Complete within a day. This is reflective-practice evidence and feeds your next Starting point.</i>                                                        |                                                                                                                                                |                                                                                                                                                       |
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| <b>What worked — and how do I know?</b><br>Calculating by hand first made the code make sense; probability debate engaged quieter maths-strong students. 75% had worms popping by the end. | <b>What will I change next time?</b><br>Show the off-by-one error deliberately next time. Give a skeleton to all students to save typing time. | <b>Students needing follow-up:</b><br>Students whose worms don't hide (CSS class issue) - approx. 3 per class; peer mentors assigned for next lesson. |

| COVERAGE CHECK <i>Tick where each must-have happens (1 · 2 · 3 = activity, C = close, H = homework). An empty box is a gap to fix before teaching. Literacy and numeracy apply in every subject.</i> |                                                                                                                                                                                                             |                                                                                                                                                                                                                        |                                                                                                                                                                                                           |                                                                                                                                                                                                  |                                                                                                                                                                                                  |                                                                                                                                                                                                                           |                                                                                                                                                                                                                        |
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| <b>Literacy</b><br>read + write · 2-1-1-6<br><input type="checkbox"/> 1 <input type="checkbox"/> 2 <input checked="" type="checkbox"/> 3 <input type="checkbox"/> C <input type="checkbox"/> H       | <b>Numeracy</b><br>numbers / data · 2-1-1-6<br><input checked="" type="checkbox"/> 1 <input type="checkbox"/> 2 <input type="checkbox"/> 3 <input checked="" type="checkbox"/> C <input type="checkbox"/> H | <b>Digital</b><br>students use it · 2-1-1-4<br><input checked="" type="checkbox"/> 1 <input checked="" type="checkbox"/> 2 <input type="checkbox"/> 3 <input type="checkbox"/> C <input checked="" type="checkbox"/> H | <b>Digital research</b><br>trusted sources · 2-1-1-4<br><input type="checkbox"/> 1 <input type="checkbox"/> 2 <input type="checkbox"/> 3 <input type="checkbox"/> C <input checked="" type="checkbox"/> H | <b>Real-life ①</b><br>application · 2-1-1-5<br><input type="checkbox"/> 1 <input type="checkbox"/> 2 <input checked="" type="checkbox"/> 3 <input type="checkbox"/> C <input type="checkbox"/> H | <b>Real-life ②</b><br>application · 2-1-1-5<br><input checked="" type="checkbox"/> 1 <input type="checkbox"/> 2 <input type="checkbox"/> 3 <input type="checkbox"/> C <input type="checkbox"/> H | <b>Inquiry</b><br>experiment / research · 2-1-1-7<br><input checked="" type="checkbox"/> 1 <input type="checkbox"/> 2 <input type="checkbox"/> 3 <input type="checkbox"/> C <input type="checkbox"/> H                    | <b>Thinking</b><br>problem-solving · 2-1-1-7<br><input checked="" type="checkbox"/> 1 <input checked="" type="checkbox"/> 2 <input type="checkbox"/> 3 <input type="checkbox"/> C <input type="checkbox"/> H           |
| <b>Create</b><br>something new · 2-1-1-7<br><input type="checkbox"/> 1 <input checked="" type="checkbox"/> 2 <input type="checkbox"/> 3 <input type="checkbox"/> C <input type="checkbox"/> H        | <b>Discussion</b><br>expressing ideas · 2-1-1-8<br><input type="checkbox"/> 1 <input type="checkbox"/> 2 <input checked="" type="checkbox"/> 3 <input type="checkbox"/> C <input type="checkbox"/> H        | <b>Pair / group</b><br>cooperative work · 2-1-1-8<br><input type="checkbox"/> 1 <input type="checkbox"/> 2 <input checked="" type="checkbox"/> 3 <input type="checkbox"/> C <input type="checkbox"/> H                 | <b>Learner choice</b><br>independent work · 3-2-1-6<br><input type="checkbox"/> 1 <input type="checkbox"/> 2 <input type="checkbox"/> 3 <input type="checkbox"/> C <input checked="" type="checkbox"/> H  | <b>Self-assessment</b><br>2-2-1-1<br><input type="checkbox"/> 1 <input type="checkbox"/> 2 <input type="checkbox"/> 3 <input checked="" type="checkbox"/> C <input type="checkbox"/> H           | <b>Peer assessment</b><br>2-2-1-1<br><input type="checkbox"/> 1 <input type="checkbox"/> 2 <input type="checkbox"/> 3 <input type="checkbox"/> C <input checked="" type="checkbox"/> H           | <b>SEN / low</b><br>specific support · 2-1-1-3<br><input checked="" type="checkbox"/> 1 <input checked="" type="checkbox"/> 2 <input checked="" type="checkbox"/> 3 <input type="checkbox"/> C <input type="checkbox"/> H | <b>Gifted</b><br>specific stretch · 2-1-1-3<br><input checked="" type="checkbox"/> 1 <input checked="" type="checkbox"/> 2 <input checked="" type="checkbox"/> 3 <input type="checkbox"/> C <input type="checkbox"/> H |

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| <b>ALWAYS VISIBLE IN DELIVERY:</b> Safe, respectful climate • every learner drawn in • purposeful use of time • smooth transitions • learner voice • specific feedback and next steps • students act on feedback |
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